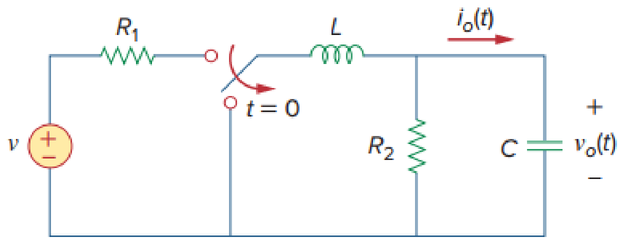
Chapter 8, Problem 25P

Problem

Using Fig. 8.78, design a problem to help other students better understand source-free *RLC* circuits.

Figure 8.78:



Step-by-step solution

Step 1 of 10

Consider the given circuit is,

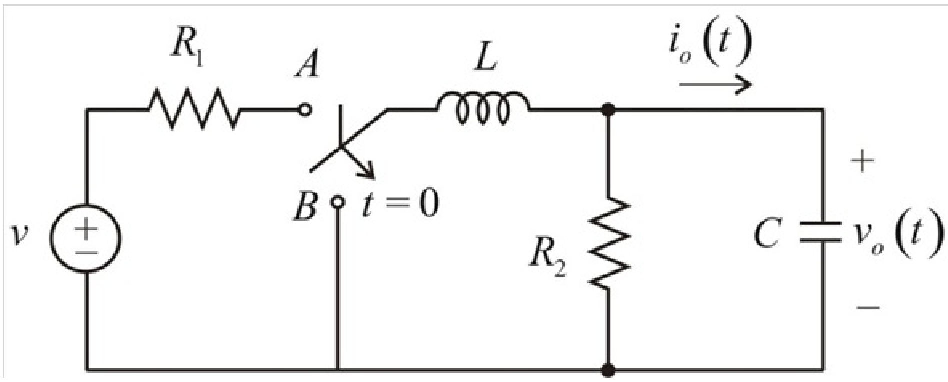
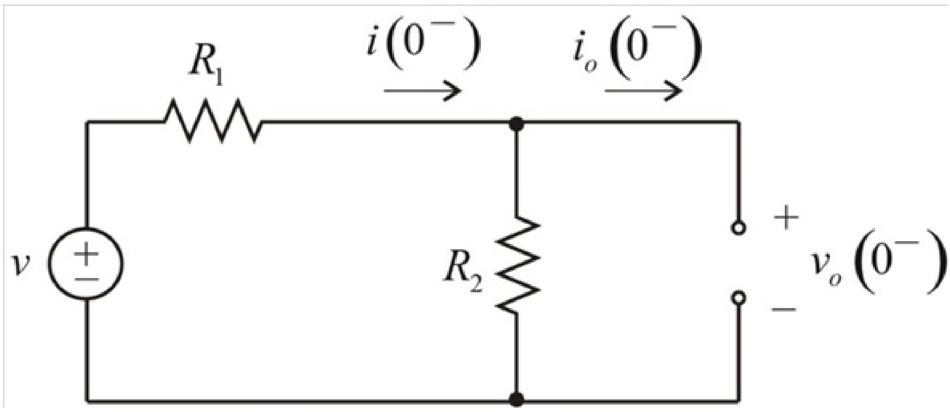


Figure 1

Step 2 of 10

For $t < 0$, the switch is in position '*A*'. The circuit is in steady state. Therefore, the inductor is replaced by a short circuit and the capacitor by an open circuit as shown below.



Step 3 of 10

Figure 2

Step 4 of 10

From Figure 2, determine the initial values across inductor and capacitor for $t = 0^-$

$$i(0^-) = \frac{v}{R_1 + R_2}$$

$i_o(0^-) = 0\text{A}$ (Due to open circuit)

$$v_o(0^-) = R_2 i(0^-)$$

$$v_o(0^-) = \left(\frac{R_2}{R_1 + R_2}\right) v$$

Step 5 of 10

Due to the continuity of the capacitor voltage and inductor current, determine the values of inductor and capacitor for $t = 0^+$.

$$i(0^+) = i(0^-)$$

$$i(0^+) = \frac{v}{R_1 + R_2}$$

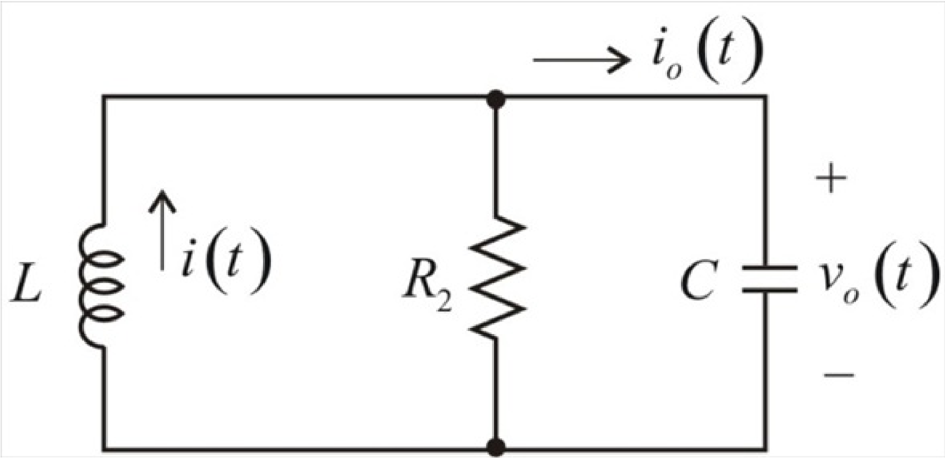
$$v_o(0^+) = v_o(0^-)$$

$$v_o(0^+) = \left(\frac{R_2}{R_1 + R_2}\right) v$$

Step 6 of 10

For $t > 0$, the switch is moved to position *B*, draw the equivalent circuit.

Step 7 of 10



Step 8 of 10

Figure 3

Step 9 of 10

Since, it is a source free parallel *RLC* circuit. Determine the value of α and ω_o.

$$\alpha = \frac{1}{2R_2C}$$

$$\omega_o = \frac{1}{\sqrt{LC}}$$

If the roots are $s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_o^2}$

There are three types of solutions,

1. If $\alpha > \omega_o$, overdamped response

$$v_o(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

2. If $\alpha = \omega_o$, critically damped response

$$v_o(t) = (A_1 + A_2 t) e^{-\alpha t}$$

3. If $\omega_o > \alpha$, underdamped response

$$v_o(t) = e^{-\alpha t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t)$$

Where $\omega_d = \sqrt{\omega_o^2 - \alpha^2}$

Where

A_1, A_2, B_1 and B_2 are constants which can be found by applying the initial conditions.

Step 10 of 10

Determine the value of output current.

$$i_o(t) = C\frac{dv_o(t)}{dt}$$

Therefore, design of RLC is completed.

There can be multiple answers for this design problem.